# Vibes / AuraLink

**Electrical Team Members**

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# How to create a reprogrammable drum pad with XYZ sensing and aftertouch?

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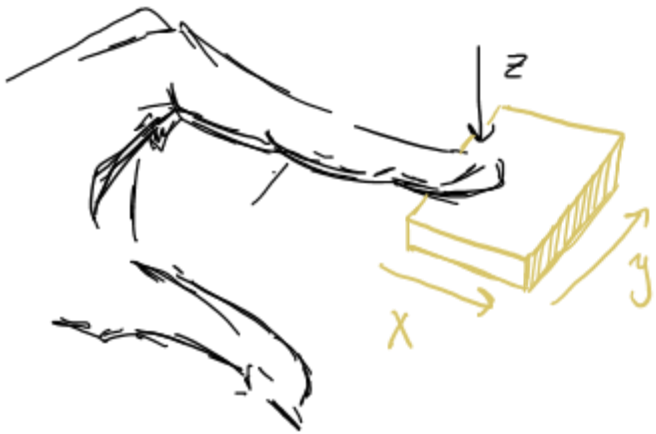
# Design Concept

- Finger drumming pad with XYZ sensing and aftertouch
  - The X-Axis
    - Calculates an internal X-coordinate from 0 to 127
    - Maps this coordinate to **MIDI Pitch Bend**
  - The Y-Axis
    - Calculates an internal Y-coordinate from 0 to 127
    - Maps it to **MIDI CC74**
  - The Z-Axis
    - **Velocity:** The initial strike determines the NoteOn velocity (1 to 127)
    - **Channel Pressure:** If you press and hold the pad, the continuous varying force is sent as sustained Channel Pressure (0 to 127).




- MIDI enabled
  - Easy integration with software team's application
  - Ability to function as a standard MIDI controller

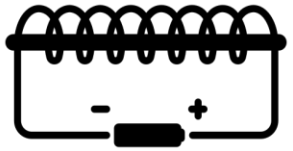
# Brainstorming

## Hall Effect

- Mechanical keyboard switch with magnet in the stem
- Use hall effect IC to generate current proportional to force from switch
- Does not allow XY sensing
- Complex and bulky

## Solenoid

- Stem through a solenoid (metal coil)
- Generates a current when user presses stem
- Provides tactile feedback
- Pushes switch back up like a spring
- Slow response time

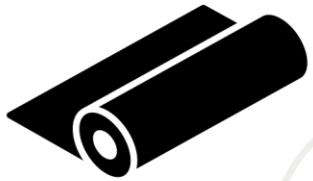


## Capacitive Grid

- Similar to a touch screen on smartphones
- Detects capacitance from user presses
- Good XY sensing
- Not robust to repeated hard hits from drumming
- Struggles at Z-Sensing

## Velostat

- Resistive fabric between copper
- Voltage generated upon user press
- Great aftertouch capabilities
- Surface material can be any insulator
- Best option to capture XYZ sensing



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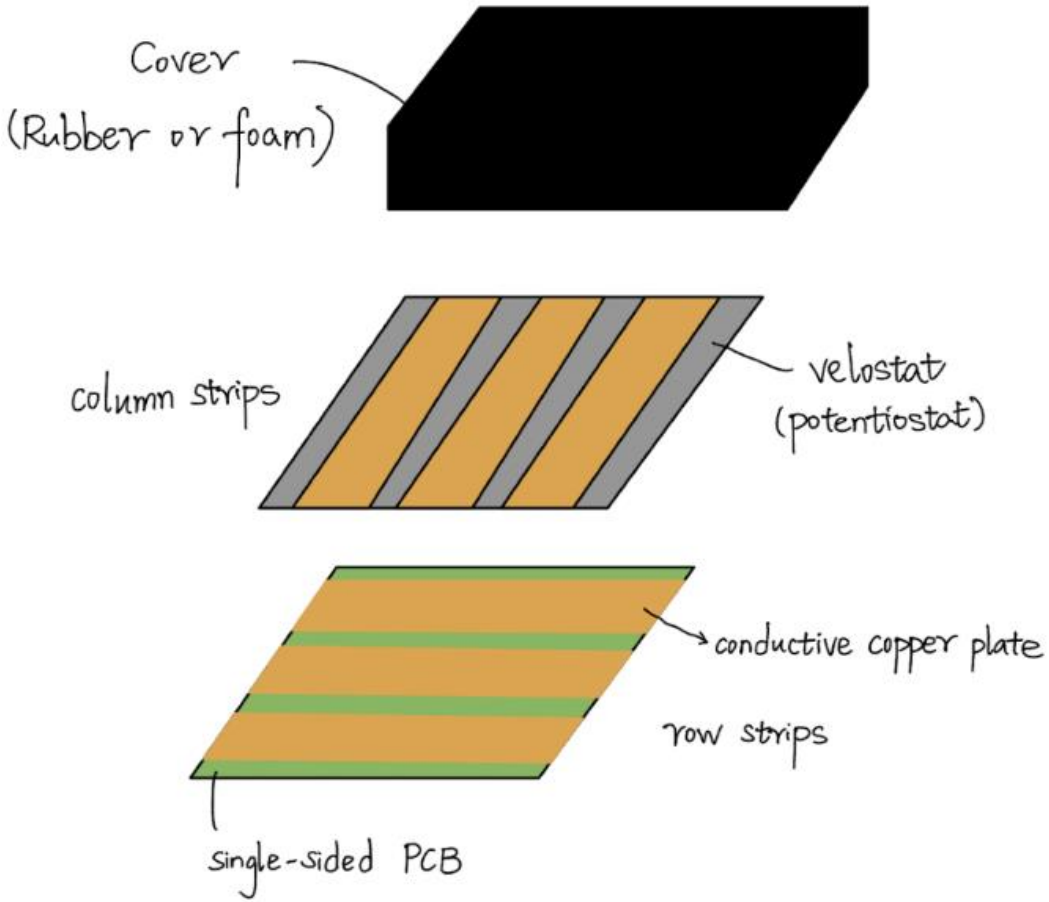
# Sensing Iterations

- Button cover, Copper tape
- Velostat (Grid or Regions)
- Copper tape, bottom piece




Both iterations offer XYZ sensing

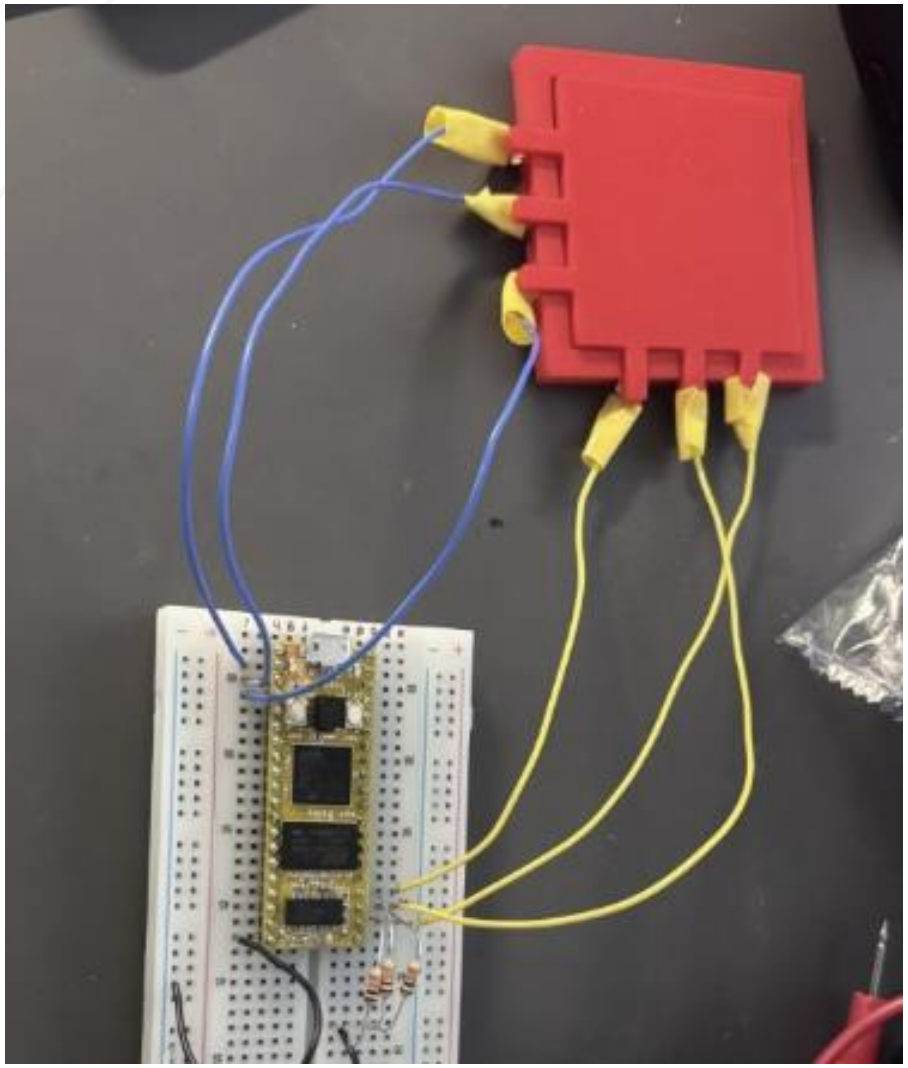
Matrix

- Optimal for high resolution XY

Regions

- Simpler software

# Prototype 1



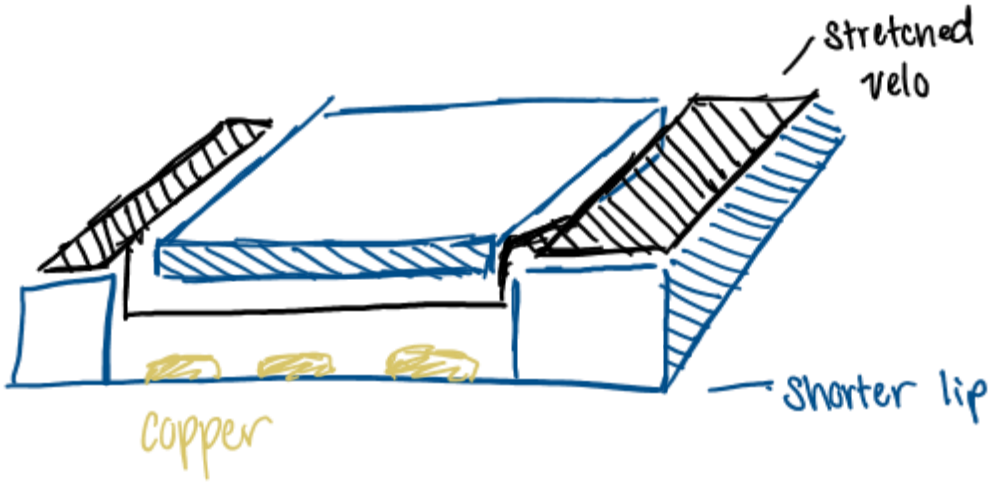
- PLA housing
- Theoretical XYZ sensing possible

Execution:

- Problems with bouncing and sticking
- Durability issues (tape kept ripping)
- Far too noisy

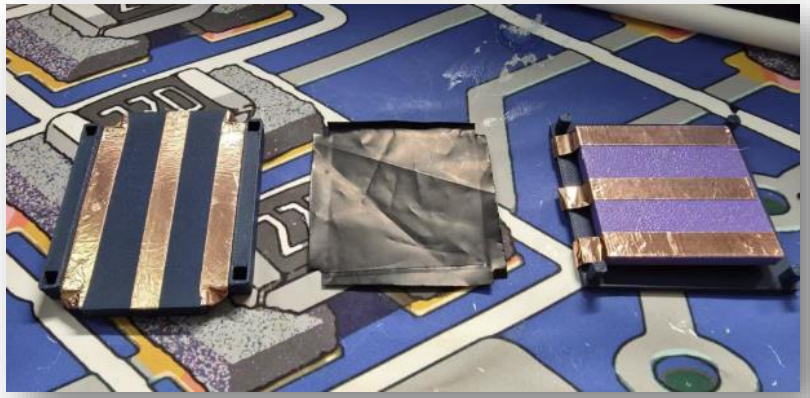
# Prototype 2

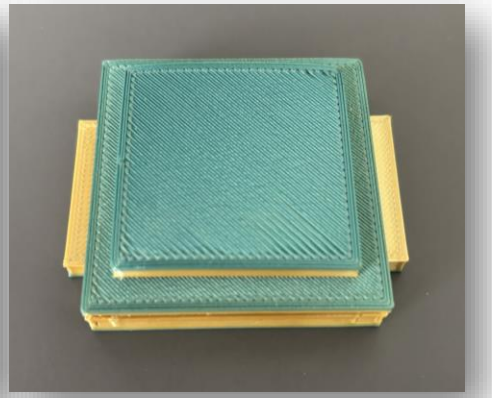
- Soft top (silicone/TPU)
- Redesigned housing
- Added top plate/mounting mechanism
- Overhauled + upgraded software




Execution:

- Produced sound in Vital
- Achieved uniform rail-to-rail sensor readings
- Fixed bouncing/sticking issues
- Functional MIDI XY control

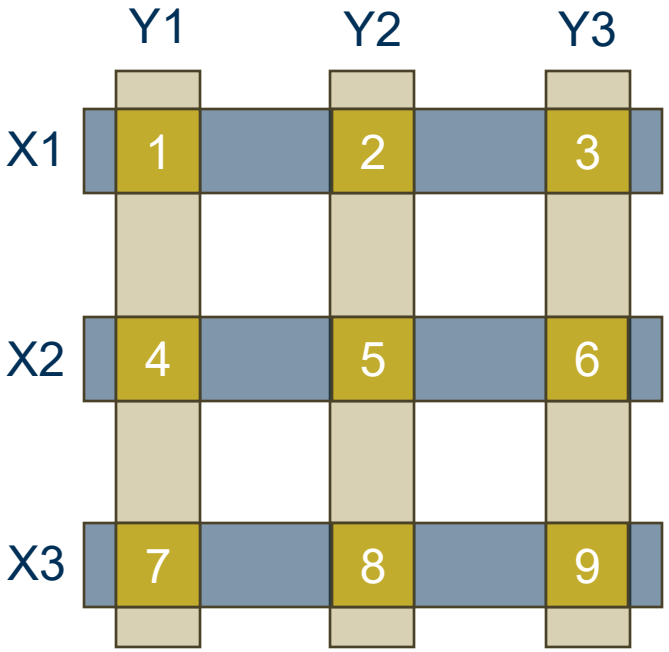


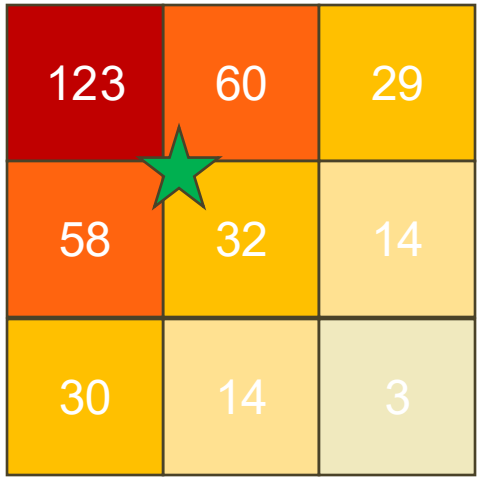


# How it Works - Electrical

- 3 Drive traces (X1, X2, X3)
- 3 Data traces (Y1, Y2, Y3)
- Multiplexed scanning

1. Set X1 HIGH, X2/X3 to Z (not GND!)
2. Read Y1, Y2, Y3
3. Set X2 HIGH, X1/X3 to Z
4. Read Y1, Y2, Y3
5. Set X3 HIGH, X1/X2 to Z
6. Read Y1, Y2, Y3



# How It Works – Firmware Pipeline

- **Signal Acquisition:** 300µs RC settling delay + First-order IIR Low-Pass Filter.
- **Dynamic Baseline Tracking:** Continually absorbs mechanical creep and temperature drift.
- **5-Point Calibration:** Maps physical boundaries and establishes a custom `VelocityMax` ceiling.
- **Weighted Centroid Algorithm:** Calculates precise, high-resolution sub-grid X/Y coordinates.
- **MPE Output:** Maps X to Pitch Bend, Y to CC74 (Timbre), and Z to Velocity/Channel Pressure.

# Pitfalls & Takeaways

- **LibDaisy via Cmake is way more powerful than daisyDuino**
  - **Access to STM32 HAL**
  - **Fully functional on USB Host / Midi / Serial**
- **ADC Resolution Mismatch:** 10-bit (Arduino) vs. 12-bit (Daisy/STM32) completely broke logic thresholds (x4 scale difference).
- **The Blocking Delay Trap:** Arduino’s slow, blocking `analogRead()` accidentally masked hardware settling times. Daisy's DMA reads required explicit delays.
- **Physics vs. Code:** "Smart" dynamic thresholds can cause system deadlocks; physical properties require hardcoded floors.

# Demo 1 – Serial Plotting

# Demo 2 – Sound!

# Next Steps

- Scaling to a Multi-Pad Architecture
  - Expanding the single 3x3 matrix into a multi-pad grid for finger drumming
  - Downsizing buttons for compatibility with cyberdecks (must be portable!)

- Vibe Patching Integration
  - The system builds custom patch in the background from user prompts and instantly assigns it to a specific zone on our Velostat matrix.
  - The moment the Daisy Seed registers a physical strike, it triggers that newly generated patch, turning a typed intention into a playable, pressure-sensitive drum voice in seconds.

# Bib

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Sean. “How to Setup Pressure Sensitive Fabric with Velostat and Conductive Fabric.” *YouTube*, 24 Sept. 2016, www.youtube.com/watch?v=8SQOBF0_80Y.

Wilson, Tracy V., Nathan Chandler, Wesley Fenlon, and Bernadette Johnson. “How the iPhone Works.” *HowStuffWorks.com*, 20 June 2007, electronics.howstuffworks.com/iphone.htm.

Firmware + CAD files: https://github.com/nayanramam/bio_patch/tree/electrical

# Thank You

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